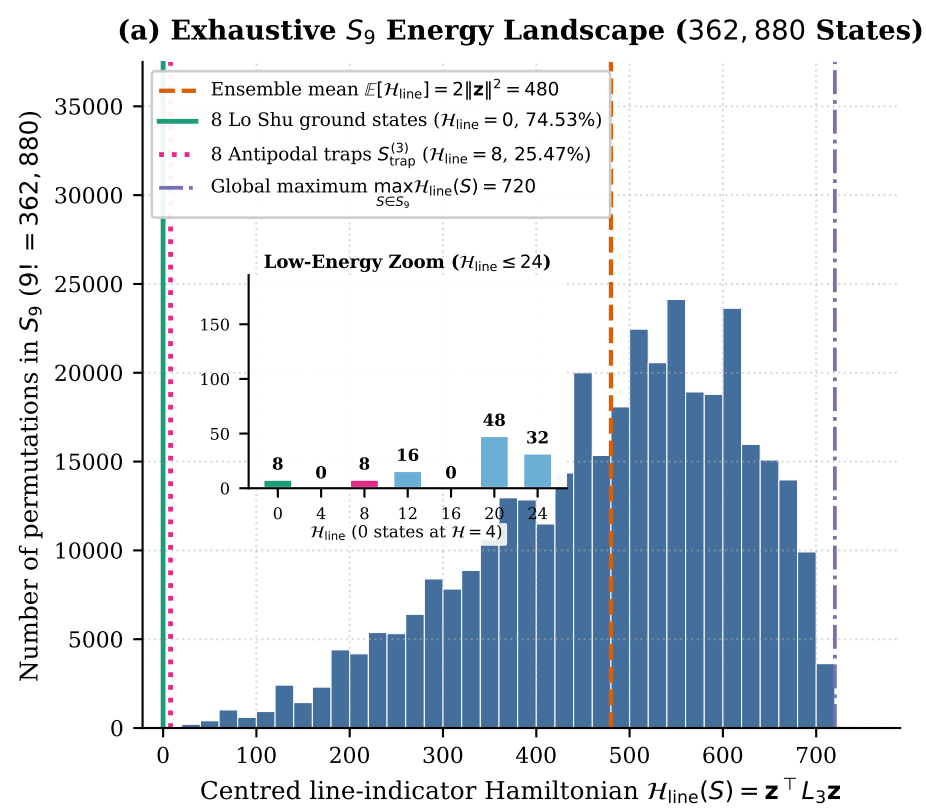
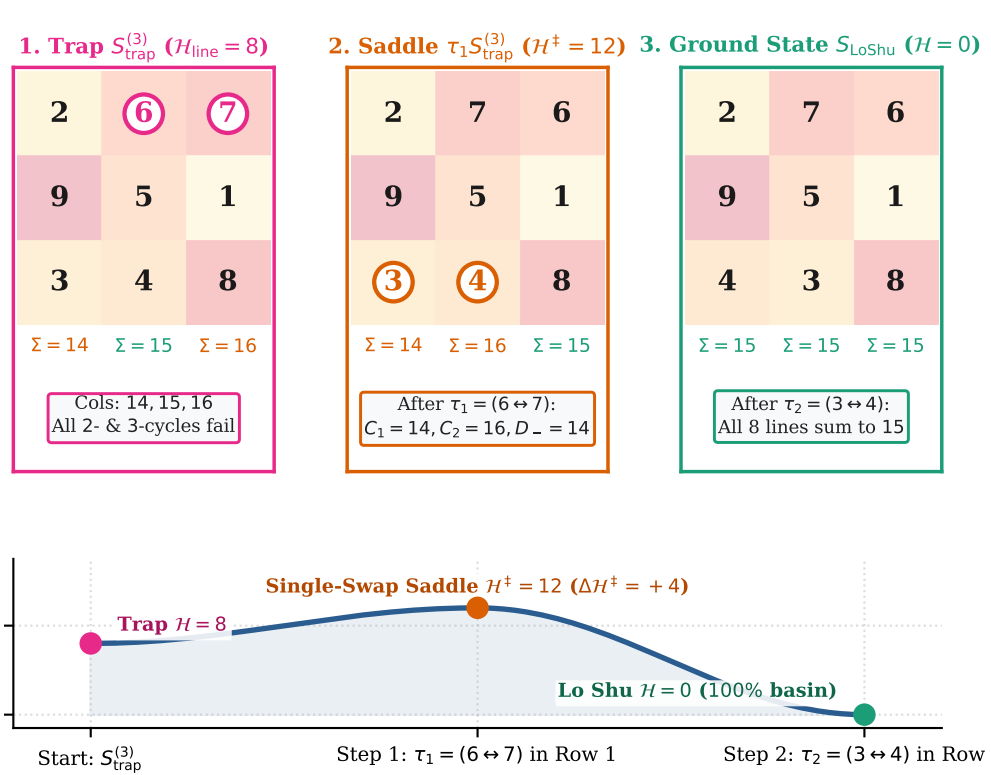

(a) Exhaustive S_9 Energy Landscape (362,880 States)
Ensemble mean $\mathbb{E}[\mathcal{H}_{\text{line}}] = 2\|\mathbf{z}\|^2 = 480$
8 Lo Shu ground states ($\mathcal{H}_{\text{line}} = 0$, 74.53%)
8 Antipodal traps $S^{(3)}_{\text{trap}}$ ($\mathcal{H}_{\text{line}} = 8$, 25.47%)
Global maximum $\max_{S \in S_9} \mathcal{H}_{\text{line}}(S) = 720$
Low-Energy Zoom ($\mathcal{H}_{\text{line}} \le 24$)
8
0
8
16
0
48
32
$\mathcal{H}_{\text{line}}$ (0 states at $\mathcal{H} = 4$)
Number of permutations in S_9 ($9! = 362,880$)
Centred line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S) = \mathbf{z}^\top L_3 \mathbf{z}$


1. Trap $S^{(3)}_{\text{trap}}$ ($\mathcal{H}_{\text{line}} = 8$)
2 6 7
9 5 1
3 4 8
Σ = 14 Σ = 15 Σ = 16
Cols: 14, 15, 16
All 2- & 3-cycles fail
2. Saddle $\tau_1 S^{(3)}_{\text{trap}}$ ($\mathcal{H}^\ddagger = 12$)
2 7 6
9 5 1
3 4 8
Σ = 14 Σ = 16 Σ = 15
After $\tau_1 = (6 \leftrightarrow 7)$:
$C_1 = 14, C_2 = 16, D_- = 14$
3. Ground State S_{LoShu} ($\mathcal{H} = 0$)
2 7 6
9 5 1
4 3 8
Σ = 15 Σ = 15 Σ = 15
After $\tau_2 = (3 \leftrightarrow 4)$:
All 8 lines sum to 15
Energy $\mathcal{H}_{\text{line}}$
Trap $\mathcal{H} = 8$
Single-Swap Saddle $\mathcal{H}^\ddagger = 12$ ($\Delta\mathcal{H}^\ddagger = +4$)
Lo Shu $\mathcal{H} = 0$ (100% basin)
Start: $S^{(3)}_{\text{trap}}$
Step 1: $\tau_1 = (6 \leftrightarrow 7)$ in Row 1
Step 2: $\tau_2 = (3 \leftrightarrow 4)$ in Row 3